

Table 4. **What each stage yields, costs, guarantees and gets wrong.** The second column is the sense in which the literature uses “confidence” at that stage. A cited phrase is a finding of the cited work; an uncited phrase follows from the definition of the stage (Table 2). Only training (S4) changes the model itself; its families are compared in Table 3.

Stage	Confidence it yields	Cost: data; inference	Guarantee	Black-box use	Typical failure	Fit for a typed decision
S0 read-off	the model’s own probability for an option or a true/false judgement [Kadavath et al. 2022]	none; none	none	no: needs token probabilities	calibration degrades after post-training [OpenAI 2023]	natural when the options are enumerated and scored [Kadavath et al. 2022]
S1 post-hoc map	a rescaled, binned or pooled probability [Guo et al. 2017], a probe’s probability from the hidden states [Azaria and Mitchell 2023], or an auxiliary model’s estimate from the input and output text [Ulmer et al. 2024a]	held-out data (labels usually; none for a fixed map); negligible	binning: distribution-free [Gupta and Ramdas 2021]; scaling: none	partly: a map needs the model’s scores and a probe its hidden states; an auxiliary model on the text needs neither	a map fitted for one task must be refitted, or predicted by a learned model, for a new one [Shen et al. 2024]	good for a fixed label set [Guo et al. 2017]
S2 prompt elicitation	a confidence stated in words or numbers [Tian et al. 2023]	none; none if asked with the answer, one more call if asked afterwards	none	yes	stated confidences cluster near the top of the scale [Xiong et al. 2023]	the confidence arrives as text and must be parsed
S3 sampling	agreement or semantic entropy across sampled answers [Farquhar et al. 2024; Wang et al. 2023b]	none; several sampled answers per query	none	yes	consistent but wrong answers look confident [Tan et al. 2025]	defined over free text; per-option use needs clustering
S3 conformal and risk control	a prediction set of options [Campos et al. 2024], or a threshold with bounded risk, such as an early exit [Schuster et al. 2022]	a calibration set; a threshold over one pass, or over several sampled answers for sets of generations	coverage $\geq 1 - \alpha$ [Campos et al. 2024], or a risk bound [Angelopoulos et al. 2025]	either	the guarantee lapses when exchangeability fails [Ulmer et al. 2024b]	a set returns several options, not one decision; a risk-controlled threshold returns one
S4 training	a confidence the model is trained to state or to assign, per option for a typed decision [Yaldiz et al. 2026]	outcomes or rewards; training compute	none in general; one reward-model method adds conformal bounds [Pan et al. 2026a]	no: needs weights	a calibration and accuracy trade-off [Yaldiz et al. 2026]	can train per-option probabilities directly [Yaldiz et al. 2026]